

RMarkdown

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Introduction

R Markdown lets you **combine code, output, and text** in the same document. You can write your analysis, include plots, and explain your results — all in one file — and render it into an HTML, PDF, or Word report.

This guide will help you:

1. Understand the basic structure of an `.Rmd` file
 2. Write text using **Markdown** formatting
 3. Add and run R code chunks
 4. Render the document to HTML
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R Markdown Structure

An R Markdown file has three parts:

1. **YAML header** – appears at the top between `---` lines (title, author, etc.)
 2. **Markdown text** – regular writing with formatting
 3. **Code chunks** – blocks of R code surrounded by three backticks
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Writing Text with Markdown

You can write *italic*, **bold**, and even lists:

- Item 1
- Item 2
- Item 3

You can also add links:

Visit RStudio's R Markdown site

And equations:

$$E = mc^2 \rightarrow E = mc^2$$

Code Chunks

To include R code, use a **code chunk** using this syntax:

Ends up looking like this:

```
# A simple calculation
x <- 5
y <- 10
x + y
```

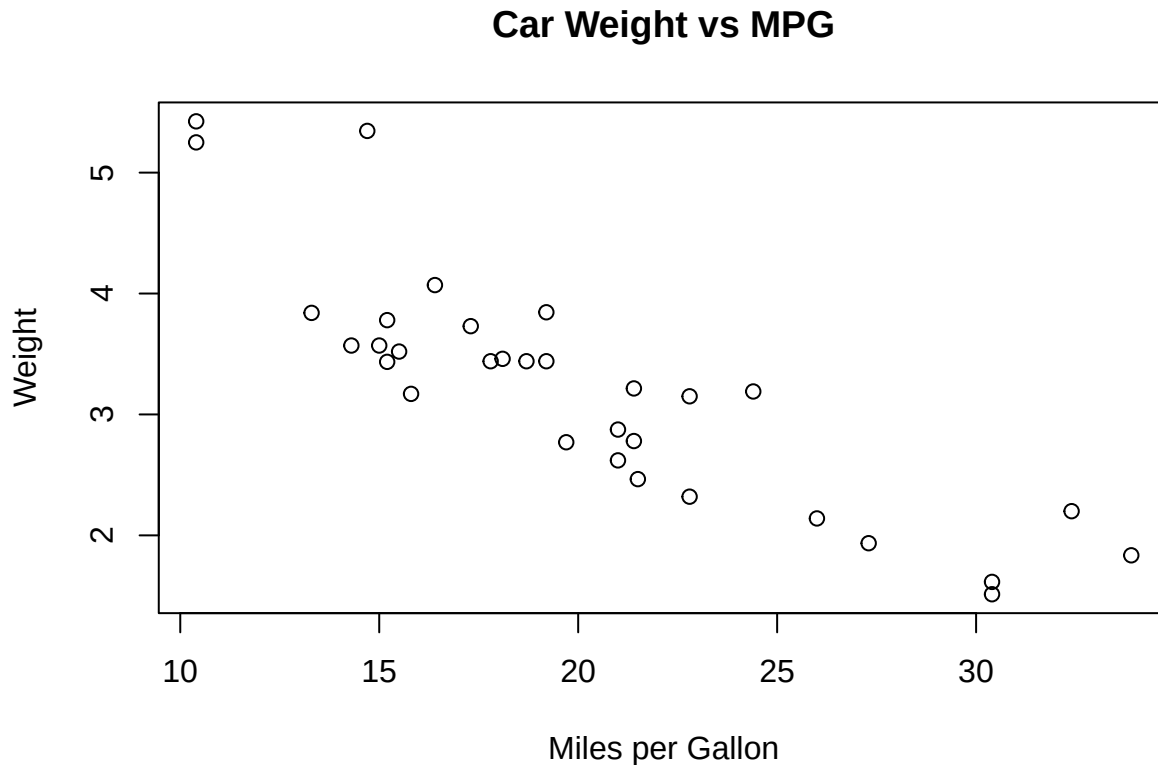
```
## [1] 15
```

Adding Plots

Plots can be added using the same 'chunk' syntax:

```
# Make a simple scatterplot

plot(mtcars$mpg, mtcars$wt,
     xlab = "Miles per Gallon",
     ylab = "Weight",
     main = "Car Weight vs MPG")
```



Chunk Options

You can control what appears in your report with chunk options in with the `{r}` curly braces: e.g.,
:

- `echo=FALSE` → hide code, show results
- `eval=FALSE` → show code, but don't run it
- `message=FALSE`, `warning=FALSE` → hide messages/warnings

Rendering Your Document

Click Knit in RStudio and choose Knit to HTML.

R will:

- Run all the code chunks
- Combine them with your text
- Produce a clean, shareable report

Tips for Success:

- Save your .Rmd file often.
- Knit regularly to catch errors early.
- Use clear headings (`#`, `##`, `###`) to organize your report.
- Keep code readable: use comments (`#`) to explain what it does.
- Load the libraries you need at the beginning of the document. Don't try to use an object you haven't defined earlier in the document. Rmd runs the R code from the beginning of the file to the end, so if the library or object haven't been defined, the Rmd file won't run.

- Don't `install.packages` or use `View(data)` in Rmd files.